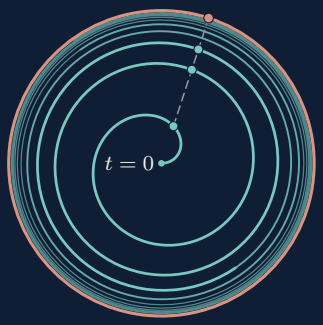
Graduate Mathematics — Winter 2027

Functional Analysis

MATH 4501 · CRN 4901

Functional analysis is the discipline in mathematics which studies spaces and algebras of functions using topology, geometry, and a modern analytic mindset. It is the foundational field for quantum mechanics, partial differential equations, operator theory, geometric analysis, and modern Fourier analysis. Thus, this is a course about infinite dimensional analysis and worlds seen only through their observables.



On the spiral $f(t) + g(t)$
On the circle $g(s)$ at phase $s \bmod 1$

At the marked phase, $g(s + n) = g(s)$.
The term f vanishes on the circle.

The Gelfand spectrum of $C_0([0, \infty)) + C(\mathbb{R}/\mathbb{Z})$

Topics

- Banach spaces
- Hahn–Banach theorem and geometric applications
- Duality and weak topologies
- Open mapping and closed graph theorems
- Hilbert spaces and bounded linear operators
- Banach algebras and C^*-algebras
- Spectral theorem for self-adjoint operators
- Unbounded operators

Instructor

Prof. Frédéric Latrémolière

Meetings

Tuesdays & Thursdays
10:00–11:50 a.m.
CMK 301

January 4–March 19, 2027